

# PIUM – Phase Interference Universe Model

## A Phase-Interference-based Model of the universe

**Unified wave-field interpretive framework for matter, mass, gravitation, detection, galactic coherence, and quantum correlation**

Version 1.3 with Extensions v1.4, v1.5, v1.6 and v1.7

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**ABSTRACT:**  
The Phase Interference Universe Model (PIUM) is a unified ontological interpretive framework in which all observable physical phenomena are understood as arising from a single universal wave field. In this model, matter is not interpreted as a collection of elementary point-like particles. What is conventionally called a “particle” is interpreted as a closed wave configuration: a spatially localized, phase-coherent, self-maintaining structure sustained by internal circulation and global phase consistency. PIUM distinguishes between open wave modes and closed wave modes. Open wave modes are propagating, non-localized configurations of  $\Phi(x,t)$  that carry phase and configuration information without forming stable material structures. Closed wave modes arise when phase relations allow for resonant, self-sustaining configurations. Such closure produces localized, persistent, and structurally stable configurations that appear as matter. Detection is not interpreted as the conversion of a wave into a particle. Instead, detection is understood as a resonant state change occurring within the material configuration of the measuring apparatus. The measuring device is itself an assembly of closed wave configurations and therefore participates in the same universal wave-field ontology as the phenomenon being measured. Mass is interpreted as internal tension within a closed wave configuration, arising from its deviation relative to the phase-neutral null-point state. This interpretation does not modify the mathematical structure of General Relativity, but does not replace established gravitational calculations. PIUM does not replace the Standard Model, General Relativity, or standard quantum mechanics. It introduces no new particles, no new fundamental forces, nor adjustable physical constants, and no alternative predictive equations. Explicit dynamical equations referenced in the extensions are presented only as compatible realizations, numerical interfaces, or phenomenological representations of phase coherence. The purpose of PIUM is to provide a coherent interpretive layer beneath existing physical theories, rather than to provide an ontological reading of matter, measurement, mass, gravitation, galactic coherence, and quantum correlation.

## 1. FOUNDATIONAL ASSUMPTION: THE UNIVERSE AS A PHASE-INTERFERENCE SYSTEM

- 1.1. PIUM assumes that the ontological basis of physical reality is a single universal wave field,  $\Phi(x,t)$ , which dynamically interferes with itself.
- 1.2. Particles, forces, and interactions are not treated as separate fundamental entities. They are interpreted as stable, phase-compatible configurations, interaction modes, and phase-organized structures of the same underlying field.
- 1.3. The global expansion of the universe is interpreted as the pre-existing evolution of interference patterns within  $\Phi(x,t)$ , not as the motion of matter through pre-large empty space.
- 1.4. PIUM introduces no metaphysical claims beyond the structural interpretation of observable physical phenomena. It is not intended as a supernatural, spiritual, or speculative metaphysical doctrine.
- 1.5. PIUM is an ontological interpretation rather than a competing predictive physical theory. Its role is to clarify conceptual relationships beneath existing formalisms while remaining constrained by established observations.
- 1.6. The central assumption is therefore not that known physics is mathematically wrong, but that the ontology beneath known formalisms can be interpreted as a single phase-interference system.

## 2. OPEN AND CLOSED WAVE MODES

- 2.1. PIUM distinguishes between open wave modes and closed wave modes.
- 2.2. Open wave modes are non-localized propagating configurations of  $\Phi(x,t)$ . They carry phase and configuration information without forming stable material structures.
- 2.3. Closed wave modes arise when phase relations allow the wave field to self-connect into a resonant, self-sustaining configuration. Such closure produces localized, persistent, and structurally stable configurations.
- 2.4. The distinction between open and closed modes is topological and phase-relational. It is not merely a distinction of energy level or local amplitude.
- 2.5. A closed mode is not a small solid object inside of matter. It is a stable phase-coherent configuration of the field itself.
- 2.6. A material object is therefore interpreted as a hierarchy of a stable closed or partially closed configurations embedded within the continuous universal field.

## 3. DETECTION AS CONFIGURATION RESPONSE

- 3.1. Detection is not interpreted as an ontological transformation of a wave into a particle.
- 3.2. Detection occurs when an incoming open wave mode induces a resonant state change within an existing closed configuration of the measuring apparatus.
- 3.3. Measuring devices are themselves assemblies of closed wave configurations. They are not external to the ontology of the measured system.
- 3.4. A measurement event is therefore a local configuration response within the detector, constrained by the phase relation between the incoming open mode and the closed structures of the apparatus.
- 3.5. This interpretation converts the operational content of measurement while avoiding the claim that matter literally becomes a particle at the moment of observation.
- 3.6. PIUM does not alter standard measurement predictions. It provides an ontological reading of why a detector registers localized events within a wave-based framework.

## 4. MASS AS INTERNAL TENSION

- 4.1. Mass is interpreted as internal tension within a closed wave configuration.
- 4.2. This internal tension arises from deviation relative to the phase-neutral null-point state.
- 4.3. Massive objects are therefore not defined by the presence of a small irreducible object, but by the persistence of a closed phase structure under internal tension.
- 4.4. This tension does not correspond to energy loss or dissipation. Closed configurations behave as lossless or quasi-lossless resonant structures.
- 4.5. Quasi-lossless modes arise from closure conditions and global phase self-consistency rather than from the assumption of discrete constituent objects.
- 4.6. The mass of a configuration reflects the structural cost of maintaining a stable deviation from the null-point reference.
- 4.7. An interpretive layer does not replace established mass-energy relations. It provides an ontological reading of why stable closed configurations appear as massive entities.

## 5. THE NULL POINT

- 5.1. The null point is defined as the phase-neutral reference state of the universal wave field.
- 5.2. The null point is not empty space.
- 5.3. It is a universal rest reference and a state of maximal latent potential.
- 5.4. Deviation from the null-point state is defined relative to this null-point reference.
- 5.5. A closed configuration persists as a structured deviation from the null point.
- 5.6. Relaxation toward the null point does not imply disappearance into nothingness. It refers to the tendency of phase-tension structures to move toward a lower-tension or more phase-neutral state.
- 5.7. The null point functions as the interpretive reference for mass, gravitation, and closure stability within PIUM.

## 6. GRAVITATION AS RELAXATION TENDENCY

- 6.1. Gravitation is interpreted as the macroscopic manifestation of the tendency of closed configurations to relax toward their own phase null-point states.
- 6.2. Closed configurations do not collapse into the surrounding wave field. They interact operationally with other phase-compatible closures and field structures.
- 6.3. Gravitational attraction reflects mutual phase compatibility and relaxation tendency rather than force mediation between fundamentally independent objects.
- 6.4. This interpretation does not modify gravitational dynamics.
- 6.5. PIUM does not alter the metric equations of General Relativity, does not change Newtonian limits, and does not introduce a new gravitational force.
- 6.6. The purpose of this interpretation is to provide an ontological account of why closed configurations exhibit gravitational behavior.
- 6.7. This interpretive layer does not replace the metric description of General Relativity remains the established predictive formalism. PIUM interprets such effective geometry as a coarse-grained representation of large-scale phase organization in  $\Phi(x,t)$ .

## 7. STRUCTURAL NON-UNIFORMITY OF SPACE

- 7.1. Because  $\Phi(x,t)$  interferes with itself, space is structurally non-uniform.
- 7.2. Interference nodes and phase-compatible regions arise where stable material organization becomes probable.
- 7.3. Large-scale structure may therefore reflect persistent interference organization within the universal field.
- 7.4. This does not imply that PIUM replaces standard cosmological structure formation models.
- 7.5. The Standard Model remains the operational framework for particle interactions.
- 7.6. The structural non-uniformity of space is therefore not an added substance or hidden medium. It is the phase-relational structure of the universal field itself.

## 8. MATTER-ANTIMATTER DISTINCTION

- 8.1. Matter and antimatter correspond to closed wave configurations that share the same null-point reference but differ in global phase orientation.
- 8.2. This distinction is not a separate wave field and does not require two independent ontological substrates.
- 8.3. Matter-antimatter annihilation corresponds to the cancellation of incompatible closure.
- 8.4. The result of annihilation is not destruction into nothingness, but a return from incompatible closed configurations to open wave modes.
- 8.5. PIUM interprets particle matter and antimatter as different orientation states of closure within a single universal field.

## 9. PARTICLES AS CLOSURE MODES

- 9.1. In PIUM, particles are interpreted as closure modes of the universal wave field.
- 9.2. Photon  
A photon is interpreted as an open wave mode. Detection corresponds to a state change within material configurations of the detector rather than a transformation of the photon into a particle-like object.
- 9.3. Neutrino and antineutrino  
Neutrino and antineutrino states are interpreted as open phase modes distinguished by orientation direction rather than as classically separable particle-antiparticle objects.
- 9.4. Electron  
The electron is interpreted as a stable fractional or spiral closure mode characterized by spin-1/2 behavior. Its phase self-identity requires a nontrivial closure relation, consistent with the need for two complete circulations in spinorial behavior.
- 9.5. Proton and neutron  
Protons and neutrons are interpreted as stable or quasi-stable composite closure configurations. Their internal phase sectors correspond to internal modes rather than separable fundamental constituents.
- 9.6. Quark-lepton structure  
In this interpretation, quarks are not treated as independently existing point-like objects detachable from the closure system. They represent internal phase sectors or organization modes within the composite closure.
- 9.7. PIUM does not deny the practical success of particle physics. It reintroduces the ontological meaning of particle descriptions as closure modes within a continuous field.

## 10. STABILITY AND PERSISTENCE

- 10.1. Closed configurations are stable due to topological constraints and global phase self-consistency.
- 10.2. Persistence arises from the self-consistency of phase relations across the entire closed configuration.
- 10.3. Stability is not produced by external confinement.
- 10.4. A closed configuration persists because its phase structure remains compatible with itself under circulation.
- 10.5. Instability occurs when the phase relations no longer support self-consistent closure.
- 10.6. Decay, transformation, and annihilation are therefore interpreted as transitions between closure states or from closed states back to open wave modes.

## 11. NON-REDUCTIONISM

- 11.1. PIUM does not reduce physical phenomena to smaller independent constituent objects.
- 11.2. Apparent constituents reflect internal organization modes rather than independent entities.
- 11.3. The reductionist model is not the ontological basis of reality in its ontology.
- 11.4. The whole configuration is not merely a sum of parts. The parts themselves are identifiable only as modes within the configuration.
- 11.5. This applies at multiple scales: particle-like structures, atoms, measuring devices, galaxies, and entangled systems are all interpreted as forms of phase organization within the same field.

## 12. RELATION TO EXISTING THEORIES

- 12.1. PIUM does not contradict the Standard Model or General Relativity.
- 12.2. It does not replace standard quantum mechanics.
- 12.3. It does not introduce new particles, forces, constants, or predictive correction terms.
- 12.4. It provides an interpretive layer beneath established formalisms.
- 12.5. The Standard Model remains the operational framework for particle interactions.
- 12.6. General Relativity remains the operational framework for gravitational dynamics.
- 12.7. Standard quantum mechanics remains the operational framework for quantum probabilities, entanglement, and measurement statistics.
- 12.8. PIUM's contribution is ontological: it asks how the successful formalisms may be understood if the underlying reality is a single self-interfering wave field.

## 13. CORE INTERPRETATIVE STATEMENTS

- 13.1. All physical phenomena arise from phase and amplitude configurations of a single universal wave field.
- 13.2. Matter is phase-coherent closure and organization, not a collection of fundamentally independent point-like entities.
- 13.3. The null point defines the universal phase-neutral reference and a state of latent potential.
- 13.4. Mass is internal tension within closed configurations.
- 13.5. Gravitation is macroscopic relaxation tendency toward null-point states.
- 13.6. Detection is resonant configuration response within the material structure of the measuring apparatus.
- 13.7. Quantum correlation is non-separable phase coherence within a globally constrained configuration.
- 13.8. PIUM provides a unified ontological reading without modifying established predictive mathematics.

## END OF PIUM v1.3 CORE FRAMEWORK

## PIUM v1.4 – MEASUREMENT CONSISTENCY AND GRAVITATIONAL VALIDATION

- 1.4.0. What PIUM does not claim  
PIUM does not introduce new forces.  
PIUM does not introduce adjustable physical parameters.  
PIUM does not modify the field equations of General Relativity.  
PIUM does not modify the Standard Model.  
PIUM does not claim predictive superiority over existing physical theories.  
PIUM proposes an ontological reinterpretation constrained by existing measurements.
- 1.4.1. Purpose of this extension  
The purpose of v1.4 is to clarify whether PIUM remains compatible with established gravitational measurements and whether it can be meaningfully confronted with observational data.  
PIUM is not introduced as a competing predictive theory to General Relativity. It is an interpretive layer beneath established gravitational formalisms.  
Such an interpretive layer remains viable only if it does not contradict well-tested measurements.
- 1.4.2. Weak Equivalence Principle  
The Weak Equivalence Principle states that all tested bodies experience identical gravitational acceleration independent of composition.  
In PIUM, gravitational response couples to the existence of a stable non-null closure mode itself, rather than to the composition of the material composition.  
Universality of free fall is therefore interpreted as the common relaxation behavior of closed configurations relative to the phase null-point reference.  
No equivalence principle test currently contradicts this interpretation.
- 1.4.3. Classical orbital dynamics  
PIUM does not alter effective spacetime geometry at macroscopic scales.  
Metric structure is interpreted as a coarse-grained representation of large-scale phase organization in  $\Phi(x,t)$ .  
Newtonian and post-Newtonian gravitational tests remain satisfied because PIUM does not modify their predictive equations.
- 1.4.4. Gravitational waves as phase-coherence measurements  
Gravitational-wave detectors measure coherent phase disturbances.  
PIUM interprets gravitational waves as propagating phase-coherence disturbances within the universal wave field.  
Observed waveforms, polarization behavior, propagation speed, and energy transport remain described by established gravitational-wave physics.  
PIUM changes the ontological reading, not the measured signal.
- 1.4.5. Distinctive measurement perspective  
PIUM suggests that gravitational events may be interpreted as large-scale phase-coherence disturbances followed by relaxation.  
This does not require modified waveforms.  
It does not require a purely absorptive and analytical direction: examining phase coherence, relaxation structure, and post-event consistency in existing datasets.
- 1.4.6. Measurement viability summary  
PIUM does not violate the Weak Equivalence Principle.  
PIUM preserves Newtonian and relativistic gravitational dynamics.  
PIUM aligns naturally with wave-based gravitational detection.  
No current measurement excludes PIUM as an ontological interpretive layer.  
Detection is interpreted as composition-dependent gravity. Intrinsic vacuum decoherence, or required violation of established gravitational propagation constraints would challenge PIUM's assumptions.
- 1.4.7. Status statement  
PIUM v1.4 is a measurement-consistent interpretive extension.  
Its strength lies in conceptual coherence, not equation replacement.
- 1.4.8. Empirical connections and future tests  
Possible empirical interfaces include:
  1. Hemispherical power asymmetry in the cosmic microwave background, interpreted as a possible residual large-scale phase-gradient structure.
  2. Gravitational waves: phase coherence, interpreted as a testable domain for relaxation and precision analysis.
  3. Precision gravitational tests, which constrain any acceptable interpretation of gravitational behavior.  
These connections are not presented as confirmations. They are proposed as future analysis pathways.
- 1.4.9. Engagement statement  
PIUM is presented as an open interpretive framework.  
Critical analysis, contradiction, and empirical challenge are explicitly welcomed.  
The model is intended to evolve through engagement while remaining under the authorship and textual control of its author.

## END OF PIUM v1.4

## PIUM v1.5 – EMPIRICAL INTERFACES AND VALIDATION PATHWAYS

- 1.5.0. Purpose of this extension  
PIUM v1.5 introduces no new postulates and modifies none of the interpretive claims of v1.3 and v1.4.  
Its purpose is to identify empirical and numerical interfaces where PIUM may be confronted with data, simulations, or quantitative analysis.  
This extension functions as:  
a structured map of falsifiable engagement points,  
a bridge between ontological interpretation and numerical work,  
and an invitation for independent analysis using existing datasets.  
Validation here means compatibility, tension, or falsification. It does not mean parameter fitting or confirmation by interpretation alone.
- 1.5.1. Cosmic Microwave Background: phase coherence at large scales  
Question:  
Do observed large-scale CMB anomalies admit a consistent interpretation as residual phase structure in  $\Phi(x,t)$ ?  
Existing data:  
Low-multipole temperature and polarization data from Planck and related CMB datasets.  
Simulation pathway:  
Generate ensembles of initial conditions with standard random phase distributions and partially correlated phase distributions consistent with a single global null-point reference.  
Evolve both ensembles using identical transfer functions.  
Quantitative analysis may include:  
multipole vector alignment statistics,  
hemispherical power asymmetry measures,  
low-temperature-polarization cross-correlation,  
and comparison against standard random-phase ensembles.  
PIUM-consistent outcome:  
If correlated initial conditions reproduce certain observed alignments without introducing new amplitudes or parameters.  
Falsification condition:  
If large-scale anomalies vanish under improved foreground separation or are statistically inconsistent with any phase-correlated ensemble, the PIUM interpretation in this domain is weakened.
- 1.5.2. Gravitation as relaxation tendency: local and classical tests  
Question:  
Is the PIUM interpretation of gravitation compatible with precision tests of gravitational universality and dynamics?  
Existing data:  
Weak Equivalence Principle experiments,  
orbital dynamics,  
post-Newtonian tests,  
and gravitational-wave observations.  
Simulation pathway:  
Define an effective phase-density or phase-relaxation representation whose coarse-grained behavior maps onto standard gravitational potentials.  
This effective construct is not introduced as an independent physical field with its own relaxation tendency, but as a descriptive mapping of already-established gravitational behavior.  
Quantitative analysis may include:  
recovery of Newtonian limits,  
absence of composition dependence,  
and scatter reduction in acceleration relations.  
PIUM-consistent outcome:  
Persistent coherence correlates with flat or near-flat outer rotation behavior without requiring halo-like fine tuning.  
Falsification condition:  
Any intrinsic decoherence, dispersion, or propagation inconsistency not accounted for by standard physics would constrain or challenge the interpretation.
- 1.5.3. Galactic dynamics: coherence versus dark matter phenomenology  
Question:  
Can long-lived phase coherence provide an alternative phenomenology to dark matter phenomenology at galactic scales?  
Existing data:  
Galactic rotation curves,  
baryonic Tully-Fisher relation,  
radial acceleration relation,  
and outer rotation-curve slope statistics.  
Simulation pathway:  
Numerical disk models with identical baryonic distributions but varying degrees of internal phase coherence.  
No exotic matter component is added in the interpretive model.  
Quantitative analysis may include:  
rotation-curve stability,  
correlation between coherence lifetime and flatness,  
outer-slope distribution,  
and scatter reduction in acceleration relations.  
PIUM-consistent outcome:  
Persistent coherence correlates with flat or near-flat outer rotation behavior without requiring halo-like fine tuning.  
Falsification condition:  
If coherence metrics show no relation to observed dynamics, or if the coherence parameters require tuning comparable to dark-matter halo parameterization, the interpretation is weakened.
- 1.5.4. Decoherence signatures in disturbed systems  
Question:  
Do interacting or perturbed systems exhibit measurable loss of phase-organized behavior?  
Existing data:  
Interacting galaxies,  
tidal distortions,  
kinematic asymmetries,  
and disturbed disk systems.  
Simulation pathway:  
Controlled perturbations are applied to coherent and incoherent disk models.  
Quantitative analysis may include:  
decay of rotational symmetry,  
growth of kinematic dispersions,  
loss of coherent spiral or disk modes,  
and comparison with observed disturbed systems.  
PIUM-consistent outcome:  
Observable loss of coherence coincides with dynamical irregularity.  
Falsification condition:  
If all observed irregularities are fully explained without any measurable organizational coherence effect, the PIUM morphology-dynamics interpretation is weakened.
- 1.5.5. Large-scale structure: topology beyond power spectra  
Question:  
Does phase organization leave topological signatures beyond standard two-point statistics?  
Existing data:  
Galaxy surveys,  
large-scale structure reconstructions,  
filament and void catalogues.  
Simulation pathway:  
Controlled field simulations initialized with identical power spectra but different phase-correlation structures.  
Quantitative analysis may include:  
filament connectivity,  
void morphology,  
genus statistics,  
topological invariants,  
and non-Gaussian phase correlation measures.  
PIUM-consistent outcome:  
Topological differences persist despite identical power spectra.  
Falsification condition:  
If topology is fully determined by amplitude information alone and no independent phase-structure signatures remain, the PIUM interpretation is weakened.
- 1.5.6. Gravitational wave phase structure  
Question:  
Are gravitational waves purely metric disturbances, or can they also be interpreted as phase-coherence propagation?  
Existing data:  
Observed inspiral, merger, and ringdown waveforms.  
Simulation pathway:  
No modification of waveform generation is required.  
The analysis is interpretive and diagnostic: phase evolution, coherence, and relaxation structure are examined within existing waveform data.  
Quantitative analysis may include:  
phase evolution consistency,  
absence of dispersive anomalies,  
post-merger coherence structure,  
and residual analysis.  
PIUM-consistent outcome:  
Full compatibility with observed phase evolution.  
Falsification condition:  
Any intrinsic decoherence, dispersion, or propagation inconsistency not accounted for by standard physics would constrain or challenge the interpretation.
- 1.5.7. Minimal numerical realization  
Question:  
Can a minimal, non-arbitrary numerical representation of PIUM be constructed?  
Approach:  
Define a phase field,  
closure conditions,  
and an effective relaxation mapping.  
Requirement:  
The model must recover Newtonian and relativistic limits without tuning.  
Failure condition:  
If no stable, convergent minimal model exists, or if stability requires arbitrary parameter fitting, the numerical program is weakened.
- 1.5.8. Status and invitation  
PIUM v1.5 does not assert empirical confirmation.  
It defines where confirmation, tension, or falsification must occur.  
Researchers interested in numerical cosmology, gravitational data analysis, wave-based ontology, galaxy dynamics, and quantum foundations are invited to engage critically and independently.

## END OF PIUM v1.5

## PIUM v1.6 – EXPLICIT WAVE DYNAMICS AND EMERGENT GALACTIC COHERENCE

- Purpose  
This extension introduces a minimal explicit dynamical representation of the universal wave field  $\Phi(x,t)$  while preserving the interpretive and ontological character of PIUM.  
No new particles, forces, dynamical laws, or predictive modifications are introduced.  
The purpose of v1.6 is not to replace existing physical theories. Its purpose is to provide a concrete starting point for numerical exploration of phase-closure stability and collective angular-momentum redistribution at galactic scales, using dynamical forms already familiar from established physics.
- 1.6.1. Minimal explicit representation  
The universal wave field may be represented by a complex scalar field:  
$$\Psi(x,t) = A(x,t) \exp(i\phi(x,t))$$
  
where  $A(x,t)$  denotes the amplitude envelope and  $\phi(x,t)$  denotes the phase field.  
The underlying dynamics of  $\Psi$  may be represented using minimal forms compatible with known wave physics, for example:  
linear wave dynamics,  
Klein-Gordon type dynamics,  
Gross-Pitaevskii type dynamics,  
and Schrödinger-Liouville type dynamics.  
Examples include:  
$$\nabla^2 \Psi = 0$$
  
$$(\partial_t + \nabla^2) \Psi = 0$$
  
or self-stabilizing nonlinear forms such as:  
$$\nabla^2 \Psi + \lambda |\Psi|^2 \Psi = 0$$
  
These equations are not proposed as new fundamental laws of PIUM. They are presented as compatible mathematical settings that admit phase coherence, circulation, and closed configurations.
- 1.6.2. The Modeling hydrodynamic representation, the phase field  $\phi$  satisfies a continuity-like constraint. Stability of closed configurations is enforced by the quantized holonomy condition:  
$$\oint \nabla \phi \cdot d\mathbf{l} = 2\pi n, \text{ where } n \in \mathbb{Z}.$$
  
This topological constraint prevents trivial collapse of closed configurations by enforcing global phase consistency.  
Rather than permitting unbounded local concentration or decay, phase gradients are distributed into extended vortex-like, spiral, or flux-tube structures.  
In the steady-state, source-free limit far from cores, the phase field satisfies:  
$$\nabla^2 \phi = 0$$
  
where the full dynamics remains governed by the underlying  $\Psi$ -equation.
- 1.6.3. Galactic rotation as collective phase-coherent dynamics  
In PIUM, galactic disks are not treated only as collections of weakly coupled test particles orbiting in a fixed central potential.  
They may also be interpreted as globally phase-coupled systems in which angular momentum is stored, transported, and redistributed through large-scale coherent modes.  
Spiral structures and density waves are not interpreted as persistent phase-coherent configurations.  
Instead, they are modeled as transient, non-stationary structures.  
Angular momentum may be carried non-locally through sustained phase gradients and winding configurations.  
Orbital velocities may be constrained by global resonance conditions and mode coupling, rather than by local enclosed mass alone.  
Within this framework, flat galactic rotation curves are interpreted as possible equilibrium signatures of collective angular-momentum transport.  
Effective centrifugal support is maintained by long-lived, globally coherent phase structures.  
This involves invoking additional matter components inside the interpretive model and modifying local gravitational laws.  
Phenomenology commonly associated with a characteristic low-acceleration scale may be interpreted as an emergent property of collective phase dynamics in low-acceleration regimes rather than as a fundamental acceleration threshold.
- 1.6.3. Asymptotic phase relaxation and flat rotation curves  
A disk galaxy may be interpreted as a coherent phase-organized configuration embedded in the universal wave field.  
At small radii, local coherent matter structures dominate the effective dynamics.  
At large radii, local structure weakens and the large-scale relaxation pattern of the field becomes increasingly important.  
In the outer region, the phase field may approach a quasi-stationary source-free relaxation regime:  
$$\nabla^2 \phi = 0$$
  
In an effectively disk-like geometry, the radial harmonic solution is logarithmic:  
$$\phi(r) = \ln r$$
  
An effective relaxation potential with logarithmic radial form gives:  
$$d\phi/dr = 1/r$$
  
and therefore an effective acceleration:  
$$g(r) = -1/r$$
  
Since circular velocity satisfies:  
$$v(r) = r g(r)$$
  
PIUM predicts a constant circular velocity.  
$$V(r) = \text{constant}$$
  
and therefore:  
$$V(r) = V_{\text{flat}}$$
  
This is the interpretive route by which flat outer rotation curves may arise as asymptotic relaxation signatures of coherent galactic phase fields.  
This is not presented as a complete predictive derivation. It is a controlled asymptotic interpretation that defines a possible field-based explanation for the observed flat rotation regime.
- 1.6.4. Observational interface  
If galactic rotation is governed partly by collective phase coherence, observed rotation curves should correlate with quantitative measures of global structural coherence.  
Quantitative relevance to observations includes:  
spiral pitch-angle distributions,  
arm-interarm contrast,  
low-order azimuthal Fourier mode dominance,  
longevity and regularity of logarithmic spiral patterns,  
and outer-disk structural persistence.  
Possible numerical pathways include:  
2D and 3D simulations of self-consistent phase-coherent systems,  
basic independent correlation structure,  
three-dimensional disk simulations with imposed large-scale phase perturbations,  
dominant  $m = 2$  spiral-mode initialization,  
 $\phi$ -based field evolution,  
and comparison with observed data.  
PIUM-consistent outcome:  
Possible observational comparisons include:  
rotation curves  $V(r)$ ,  
velocity dispersions,  
spiral pitch-angle statistics,  
coherence lifetimes,  
morphological Fourier decomposition,  
and matched SPARC data imaging datasets.
- 1.6.5. Falsification condition  
If flat rotation curves show no systematic correlation with spiral coherence, morphological coherence, pitch-angle structure, or other phase-organization metrics across a statistically significant sample, the phase-coherence interpretation is weakened.  
If coherence metrics show no relation to observed dynamics, or if the coherence parameters require tuning comparable to dark-matter halo parameterization, the interpretation is weakened.  
The interpretive value of this extension is also weakened.
- 1.6.6. Status and invitation  
PIUM v1.6 is not intended to replace existing physical theories.  
It does not introduce alternative gravitational dynamics.  
Its purpose is to provide a bridge toward explicit numerical realization while preserving the ontological character of PIUM.  
It is not intended as a competing predictive theory to General Relativity.

## END OF PIUM v1.6

## PIUM v1.7 – PHASE-CORRELATED CLOSURE AND QUANTUM ENTANGLEMENT

- Purpose  
This extension incorporates quantum entanglement into the ontological framework of PIUM by describing independent objects as non-separable phase-coherent configurations of the universal wave field  $\Phi(x,t)$ .  
No new particles, forces, dynamical laws, or quantitative departures from standard quantum mechanics are introduced.  
The purpose of this section is interpretive: to provide a unified account of quantum non-local correlations within the same wave-field ontology used by PIUM to describe matter, detection, gravitation, and stable closure.
- 1.7.1. Entanglement as non-separable phase-coherent configuration  
In PIUM, physical systems are not regarded as fundamentally independent entities.  
They are locally identifiable configurations within a single universal wave field.  
Entanglement is interpreted as the condition in which two or more subsystems do not correspond to fully independent closed configurations.  
Instead, they form a single non-separable phase-coherent structure within  $\Phi(x,t)$ .  
Accordingly:  
The physically relevant phase relations are defined at the level of the composite configuration rather than at the level of isolated subsystems.  
Closure conditions are jointly constrained,  
the total state is not factorizable into independent local descriptions,  
and the state of each subsystem is not ontologically complete without reference to the total configuration.
- Entanglement is therefore not interpreted as a connection established between distant, otherwise separate objects.  
It is interpreted as the persistence of an underlying extended configuration that is only operationally partitioned during observation and measurement.
- 1.7.2. Non-local correlation as global constraint  
Because the universal wave field is continuous and globally defined, the consistency conditions governing phase-coherent configurations may also be global.  
When spatially separated subsystems belong to the same non-separable closure structure, the correlations observed between measurement outcomes are interpreted as consequences of that prior global constraint.  
This interpretation, if correct, would require a non-local signal exchange mechanism.  
The observed non-local correlations do not arise from dynamical communication between independent localized objects after measurement.  
They arise because the measured subsystems were not ontologically independent components of the total configuration.
- PIUM therefore treats quantum non-locality as a manifestation of non-separability, not as a violation of relativistic causality.  
The global constraint is not directly observable as a physical signal.  
It is inferred through statistical observations across many realizations.
- 1.7.3. Measurement as local configuration selection  
PIUM interprets measurement as a resonant state change within the material configuration of the measuring apparatus.  
For entangled systems, measurement is not described as the transformation of a wave into a particle.  
It is also not described as the creation of a new physical connection between distant subsystems.  
Detection is interpreted as a local realization consistent with a globally constrained, non-separable phase configuration.
- This does not imply pre-existing definite local values independent of measurement context.  
The correlated character of distant outcomes reflects the non-separable structure of the total configuration.  
It is a post-measurement physical influence propagating between detectors.  
This formulation is intended as an ontological interpretation only.  
It does not modify the standard formal treatment of measurement in quantum mechanics.
- 1.7.4. Compatibility with quantum mechanics  
PIUM does not alter the mathematical formalism of quantum mechanics.  
It introduces no correction to established quantum predictions.  
All standard results remain unchanged, including:  
All standard quantum mechanics results,  
basis-independent correlation structure,  
non-factorizable composite-state statistics,  
and standard probability assignments for measurement outcomes.  
Within PIUM, these results are interpreted as consequences of:  
non-separable phase-coherent structure,  
globally constrained closure relations,  
and the emergent rather than fundamental individuality of subsystems.  
This section therefore does not compete with quantum theory at the predictive level.  
It proposes only an ontological reading of already established formal content.
- 1.7.5. Conceptual distinction from classical separable models  
Classical separable models generally assume that physical systems are composed of independent objects whose correlations must be mediated through local interactions or shared hidden variables.  
PIUM assumes instead:  
a single entity in a single universal wave field,  
and global non-separability as a permissible phase-consistent localization.  
Entanglement is therefore not treated as an exceptional mechanism added to otherwise independent objects.  
It is interpreted as a direct expression of the underlying ontological continuity of  $\Phi(x,t)$ .
- 1.7.6. Scope, limits, and epistemic status  
This extension is interpretive rather than dynamically novel.  
It introduces no new dynamics, particles, or predictive claims.  
It does not replace Hilbert-space methods.  
It does not solve the measurement problem in the sense of providing a new predictive collapse model.  
It does not yield new empirical predictions beyond standard quantum mechanics.  
Its viability depends on:  
internal conceptual coherence,  
consistency with established quantum results,  
and compatibility with the broader PIUM ontology.
- The interpretation would be weakened if a fully satisfactory strictly local separable account of Bell-type correlations were established.  
Conversely, the continued empirical success of non-separable quantum descriptions remains compatible with the PIUM interpretation of entanglement as global phase-coherent closure.
- 1.7.7. Status  
PIUM v1.7 is fully consistent with v1.3–v1.6.

## PIUM v1.7 – FULL DOCUMENT

## FINAL INTEGRATED STATUS STATEMENT

- PIUM is a unified ontological interpretive framework.  
It begins from a single universal wave field  $\Phi(x,t)$ .  
Matter is interpreted as phase-coherent closure.  
Mass is interpreted as internal tension relative to a null-point reference.  
Detection is interpreted as resonant configuration response.  
Gravitation is interpreted as relaxation tendency toward phase-neutral states.  
Galactic flat rotation behavior is interpreted as a possible large-scale coherence and relaxation signature.  
Quantum entanglement is interpreted as non-separable phase-correlated closure.  
PIUM does not replace the Standard Model.  
PIUM does not replace General Relativity.  
PIUM does not replace standard quantum mechanics.  
PIUM does not replace the metric description of General Relativity.  
Its purpose is to provide a coherent ontological layer beneath established formalisms, linking classical, quantum, galactic, and cosmological phenomena through the common language of phase, amplitude, closure, relaxation, and global field coherence.

## END OF FULL DOCUMENT